

1. Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that enables computers to learn from historical data and make automated predictions without explicit rule programming. It is broadly categorized into supervised learning, unsupervised learning, and reinforcement learning paradigms.

2. Supervised Learning Techniques

Supervised learning algorithms are trained using labeled datasets where the desired ground truth is known in advance. Common supervised models include linear regression, decision trees, support vector machines, and deep artificial neural networks.

3. Deep Learning and Neural Networks

Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for image classification and computer vision applications. They extract hierarchical spatial patterns through convolutional layers, pooling filters, and fully connected activation layers.

4. Real-world Impact and Conclusion

Machine learning has fundamentally transformed automated decision making across medical healthcare diagnostics, autonomous vehicles, financial fraud detection, and natural language processing systems.